

Vulnerability Assessment Report

12st August 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 2026 to August 2026. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The database server stores and manages large amounts of data that employees rely on to identify and manage potential customers. Because the server contains valuable business information, it is crucial to protect it from unauthorized access, data theft, and misuse that could compromise sensitive information and disrupt business operations. Disabling the server may reduce the risk of security incidents; however, completely shutting it down could also prevent employees from accessing essential data and cause delays in daily operations. Therefore, security measures should be implemented to protect the database server while maintaining its availability and supporting business continuity.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Hacker	Obtain sensitive information via exfiltration	3	3	9
Advanced persistent threat (APT)	Perform reconnaissance and surveillance of organization	3	3	9

<i>Customer</i>	<i>Alter/Delete critical information</i>	<i>1</i>	<i>3</i>	<i>3</i>
<i>Employee</i>	<i>Disrupt mission-critical operations</i>	<i>2</i>	<i>3</i>	<i>6</i>

Approach

Risks considered the data storage and management methods of the business. The likelihood of a threat occurrence and the impact of these potential events were weighed against the risks to day-to-day operational needs.

Remediation Strategy

Implementation of authentication, authorization, and auditing mechanisms to ensure that only authorized users access the database server. This includes using strong passwords, role-based access controls, and multi-factor authentication to limit user privileges. Encryption of data in motion using TLS instead of SSL. IP allow-listing to corporate offices to prevent random users from the internet from connecting to the database.